FOURIER TRANSFORMATION

IMAGE TRANSFORMATION

Image Transformations generate ‘new’ images from two or more sources which highlight particular features or properties of interest, better than the original input images.

NEED OF TRANSFORMATION

An image transformation can be applied to an image to convert it from one domain to another. Viewing an image in domains such as frequency enables the identification of features that may not be as easily detected in spatial domain.

All transforms will not give frequency domain information. Most of image transforms like Fourier transform, Wavelet transform etc. gives information about the frequency contents in an image transforms and allow us to extract more relevant information.

FOURIER TRANSFORMATION

INTRODUCTION

Fourier transformation introduce by Jean Baptiste Joseph Fourier. His interest of area is distribution of heat and conduction.

Image click in different heat temperature such as take photograph in sunlight if sunlight is coming from backward then image is black if the sunlight is coming from the frontend then image is bright.
It create different formation and effects of image. Fourier transformation method use for correct of this formation and effects.

DEFINATION

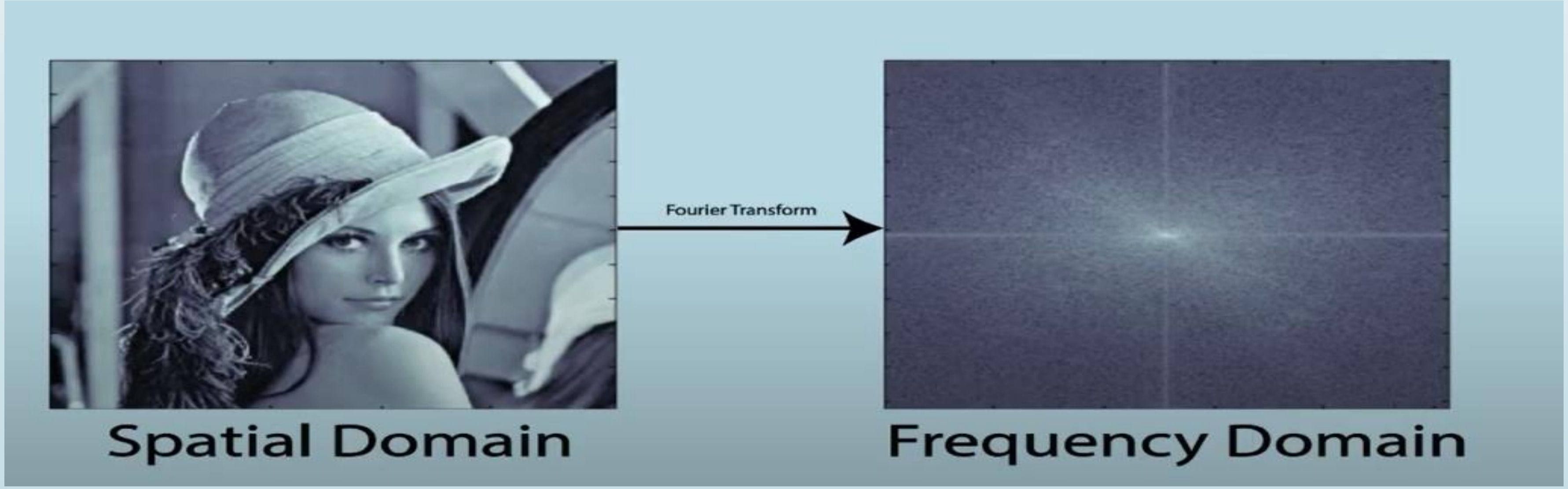
Fourier transformation convert image into spatial domain to frequency domain or frequency domain to spatial domain.

Spatial domain

The normal row/column coordinates system in which images are normally expressed is called Spatial domain

Frequency domain

The coordinates of the 2D space in which these scale components are represented are given in terms of frequency called Frequency domain



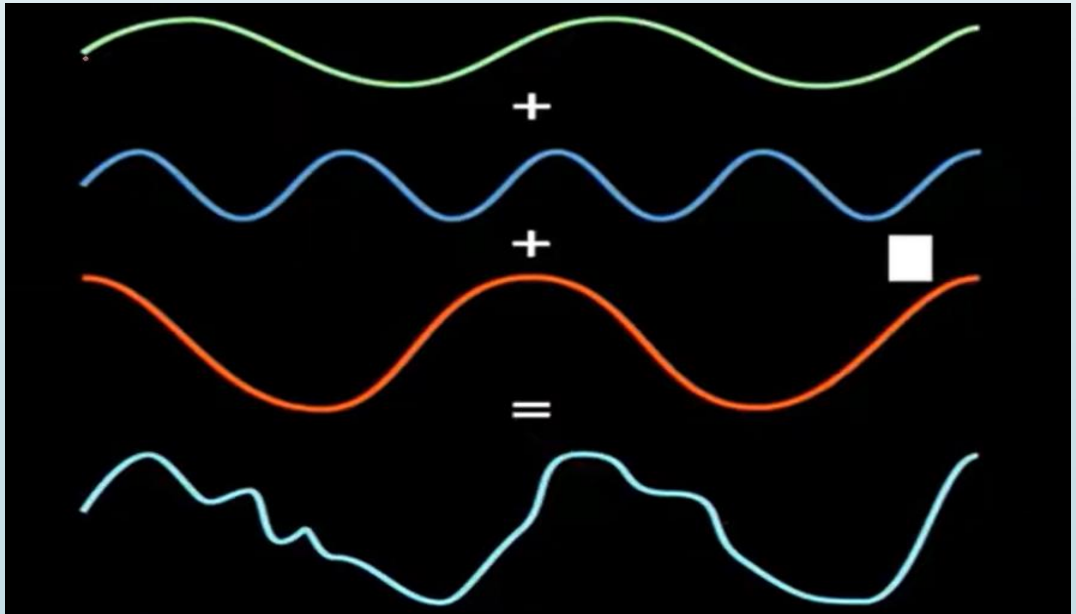
WHAT IS FOURIER TRANSFORM

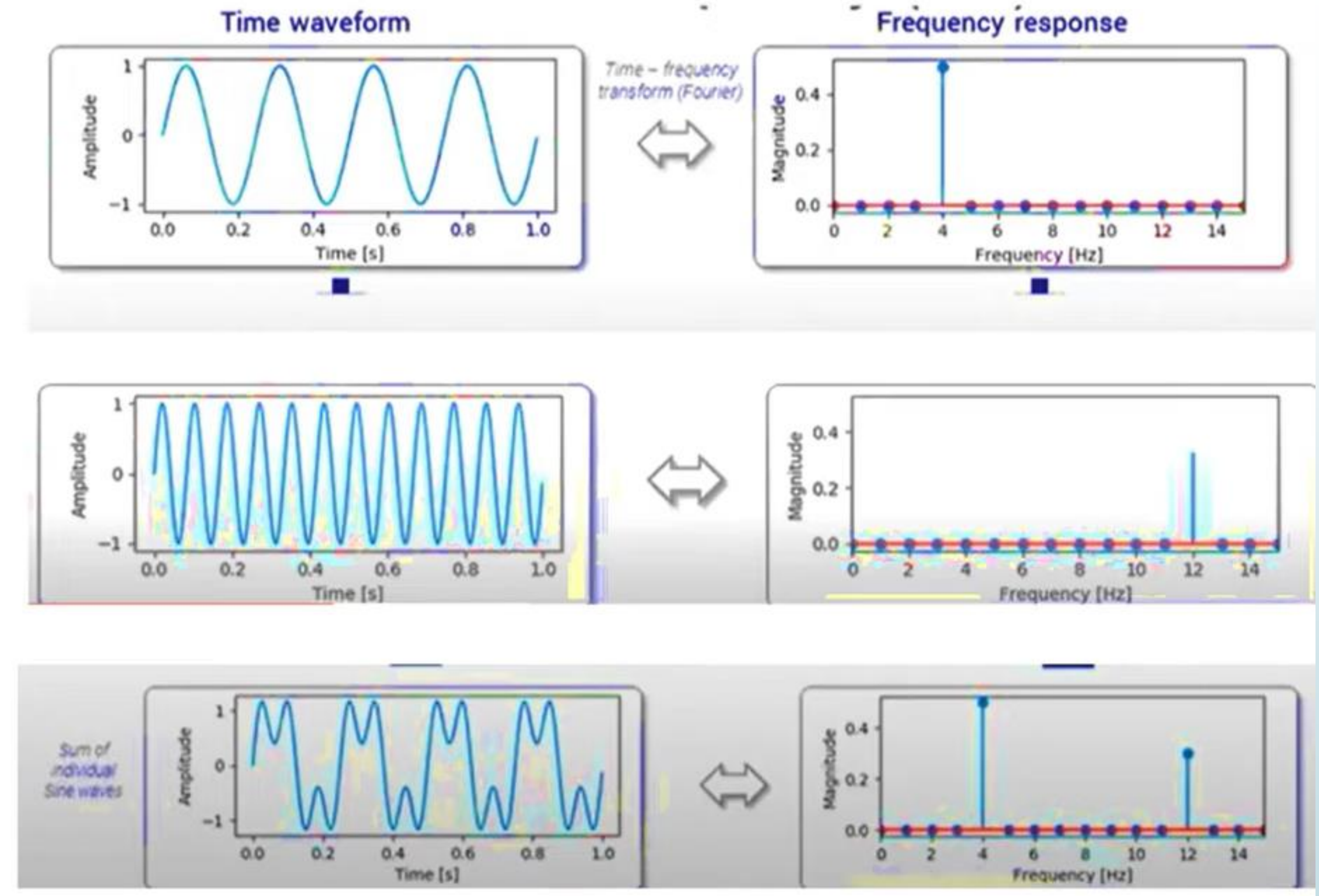
Fourier transform is a mathematical method to convert a function in the amplitude vs time domain to the amplitude vs frequency domain for non-periodic functions.

OR

Fourier transform is used to decompose an image into its sine and cosine components. The output of the transformation represents the image in the Fourier or frequency domain, while the input image is the spatial domain equivalent. In the Fourier domain image, each point represents a particular frequency contained in the spatial domain image.

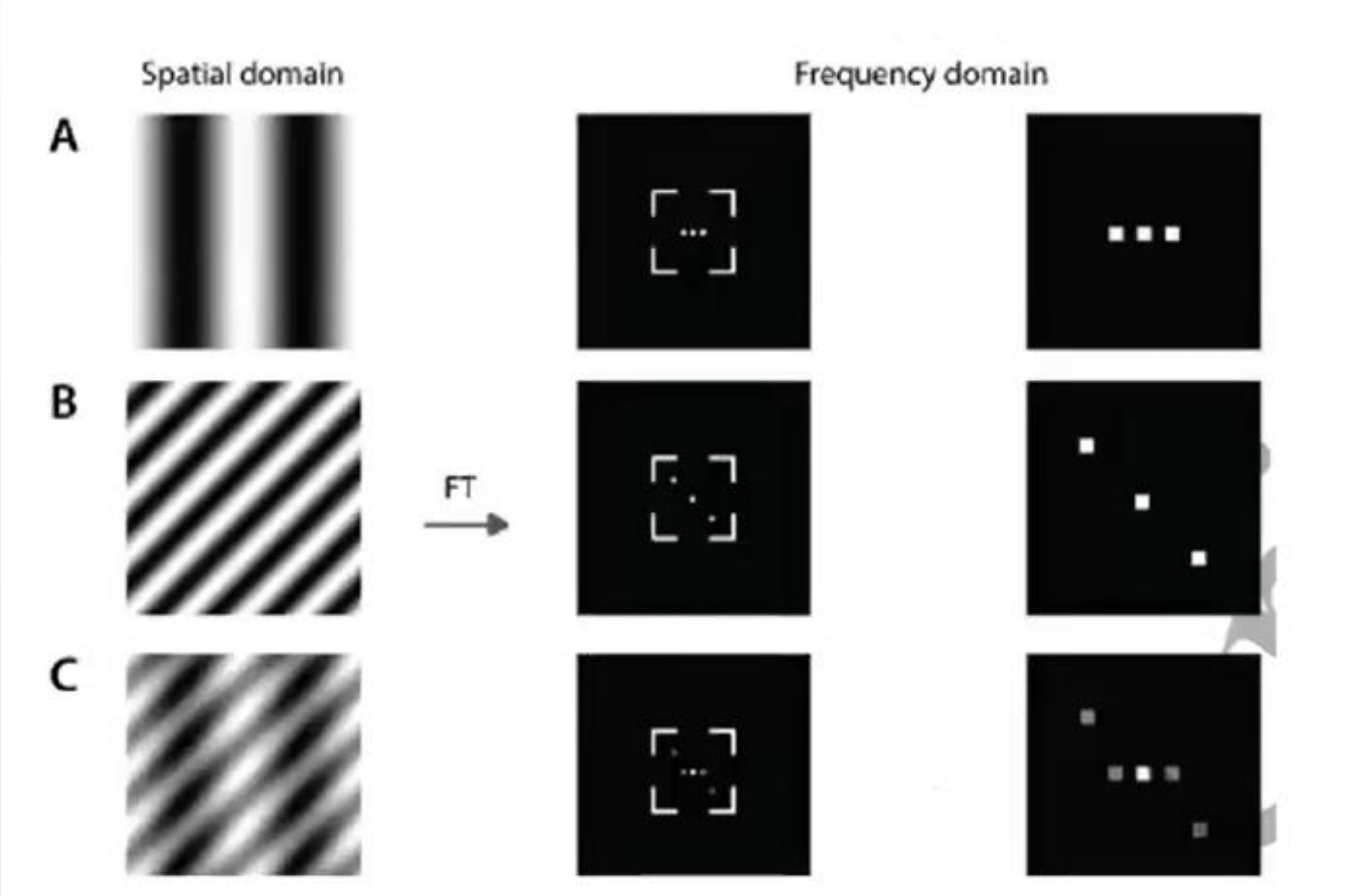
It shows that any waveform can be written as weighted sum of sinusoidal functions




Time waveform
Frequency response
Time – frequency transform (Fourier)
Amplitude
Time [s]
Magnitude
Frequency [Hz]
Sum of individual Sine waves

WHAT INFORMATION FOURIER TRANSFORM OF AN IMAGE GIVE ?

The response of the Fourier transform to periodic patterns in the spatial domain images can be seen very easily in the following images:



Types of Fourier Transformation

- Continuous Fourier Transformation
- Discrete Fourier Transformation

➢ 1-D Continuous Fourier Transformation

Fourier Transformation

Let f(x) is a continuous function of some variable then the Fourier transformation of f(x) is f(u)

$$F(f(x)) = F(u) = \int_{-\infty}^{+\infty} f(x).e^{-j2\pi ux}.dx \quad ,j = \sqrt{-1}$$

Here f(x) must be continuous

Where f(x,y) is the image in the spatial domain

Exponential term is the basic function. The basic function are sine and cosine wave with increasing frequencies.

$$\boldsymbol{e^{-j\theta} = cos\theta + j\,sin\,\theta}$$

Inverse Fourier Transformation

F(u) is a Fourier transform of signal f(x) so after inverse Fourier transformation of f(u) we get f(x)

$$F^{-1}(f(u)) = F(x) = \int_{-\infty}^{+\infty} f(u).e^{-j2\pi ux}.du$$

Where x is time in seconds, unit of u are Hz or cycles/sec

Fourier transformation pair

F(u) Fourier transformation of signal f(x)

F(x) is original signal

Here F(u) is a complex function contains real & imaginary part

$F(u) = R(u) + jI(u)$

Magnitude of Fourier transformation

$|F(u)| = \sqrt{R^2(v) + I^2(u)}$

Phase

$\varphi(F(u)) = \tan^{-1}\frac{I(u)}{R(u)}$

Magnitude Phase

$F(u) = |F(u)|e^{j\varphi(u)}$

Power of f(x)

$|F(u)|^2 = R^2(u) + I^2(u)$

2-D CONTINUOUS FOURIER TRANSFORMATION

Fourier Transformation

Let f(x,y) is 2 dimensional signal with 2 variable

$$f(u,v) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y).\, e^{-j2\pi(ux+vy)}\,.\, dxdy$$

Inverse Fourier Transformation

$$f(x,y) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(u,v).\, e^{-j2\pi(ux+vy)}\,.\, dudv$$

Spectrum

$$|F(u,v)| = [R^2(u,v) + I^2(u,v)\ ]^2$$

Phase

$$\varphi(F(u,v)) = \tan^{-1}\frac{I(u,v)}{R(u,v)}$$

Power Spectrum

$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$$

➢ DISCRETE FOURIER TRANSFORMATION

The DFT is the sampled Fourier Transform and therefore does not contain all frequencies forming an image but only a set of samples which is large enough to fully describe the spatial domain image. The number of frequencies corresponds to the number of pixels in the spatial domain image. The image in the spatial and Fourier domain are the same size.

Forward Discrete Fourier Transformation

Let we have an image of size M*N then F(u,v) is the Fourier Transformation of image f(x,y)

$$F(k) = \sum_{k=0}^{N-1} f(x)\, e^{-j2\pi kx/N} \quad , \quad 0 \leq k \leq N-1$$

Where variable u = 0,1,2,…..M-1 and v= 0,1,2,…,N-1

Inverse Discrete Fourier Transformation

$$F^{-1}(F(k)) = F(n) = \frac{1}{N}\sum_{k=0}^{N-1} f(x) \,.\, e^{-j\frac{2\pi}{N}kn} \quad , \quad 0 \leq n \leq N-1$$

Where variable x= 0,1,2,…..M-1 and y= 0,1,2,…,N-1

2D Discrete Fourier Transformation

Discrete Fourier Transformation

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)\, e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

Inverse Discrete Fourier Transformation

$$F(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} f(u,v)\, e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

u&v transform grey variables

x&y spatial image variables

PROPERTIES OF FOURIER TRANSFORMATION

- Seperability
- Periodicity
- Conjugate
- Rotation
- Distributive
- Scaling
- Convolution
- Correlation

Seperability

$$F(u,v) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} f(x,y)\, e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

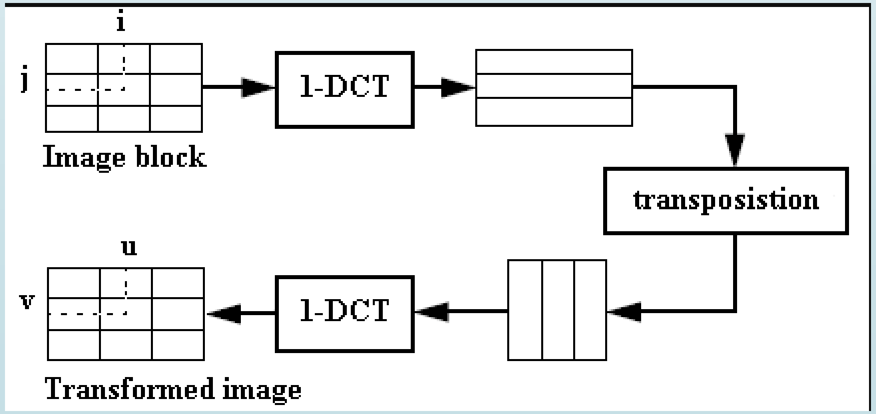
The DFT can be expressed in separable form

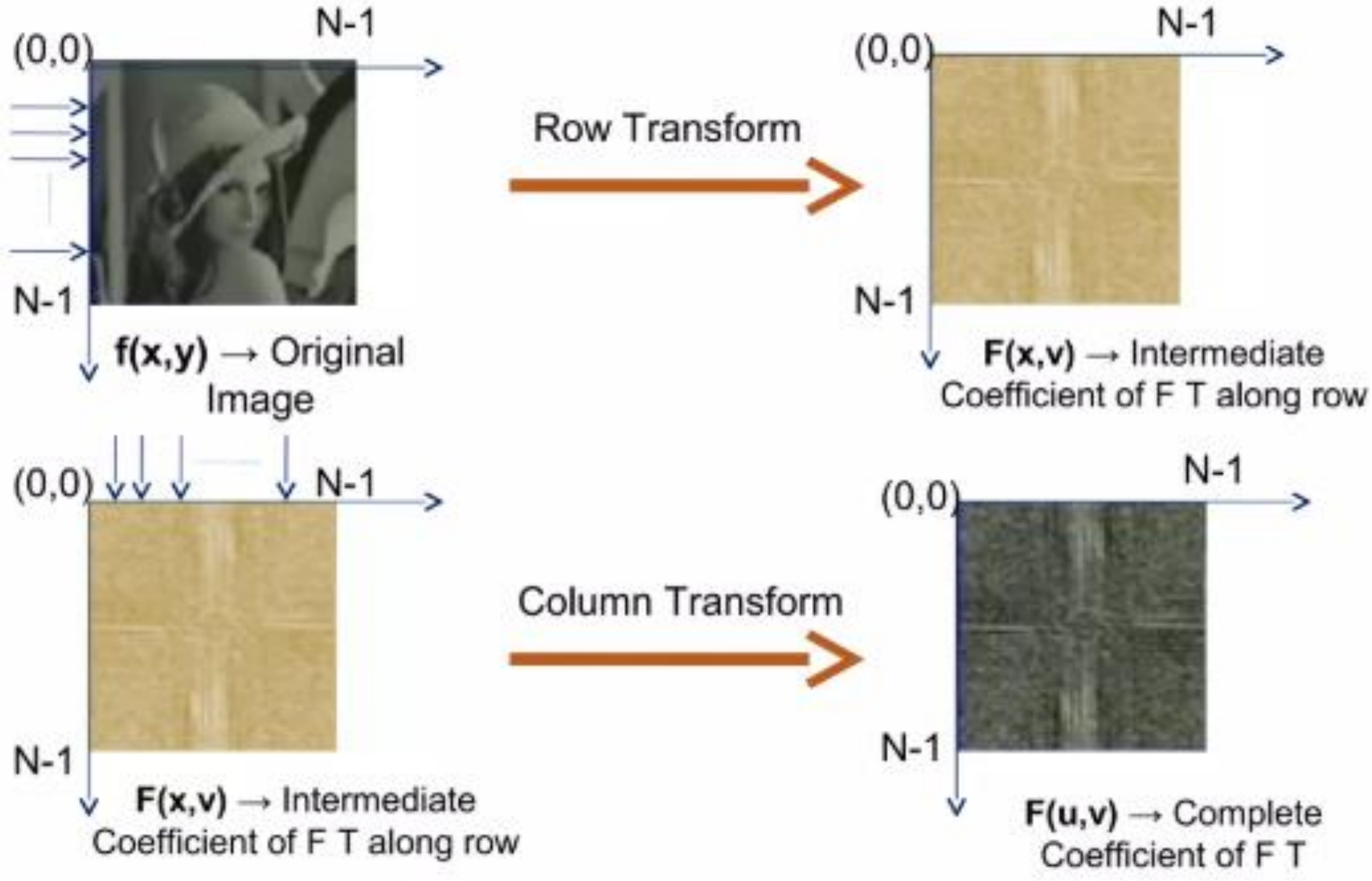
$$F(u,v) = \frac{1}{M}\sum_{u=0}^{M-1} e^{-j2\pi\frac{ux}{M}} \cdot \frac{1}{N}\sum_{v=0}^{N-1} f(x,y)\, e^{-j2\pi\frac{vy}{N}}$$

$$= \frac{1}{M}\sum_{u=0}^{M-1} F(x,v) e^{-j2\pi\frac{ux}{M}}$$

Where $F(x,v) = \frac{1}{N}\sum_{y=0}^{N-1} F(x,y)\, e^{-j2\pi\frac{vy}{N}}$

It tells us that we can compute the 2D transform by first computing 1-D transformation along each row of the input image then computing 1-D transform along each column of this intermediate result.




(0,0)
N-1
N-1
f(x,y) → Original Image
Row Transform
(0,0)
N-1
N-1
F(x,v) → Intermediate Coefficient of F T along row
(0,0)
N-1
N-1
F(x,v) → Intermediate Coefficient of F T along row
Column Transform
(0,0)
N-1
N-1
F(u,v) → Complete Coefficient of F T

Periodicity

Fourier transform and Inverse fourioer transform are periodic if M is the period

$$F(u,v) = F(u+M,v) = F(u,v+M)$$
$$=F(u+M,v+M)$$

F(x,y) are real

FT shows conjugate symmetery

$$F(u,v) = |F(u,v)| = |F(-u,-v)|\, F^*(-u,-v)$$
$$\& \; |F(u,v)| = |F(-u,-v)|$$

Periodicity property says that the Discrete Fourier Transform and Inverse Discrete Fourier Transform are periodic with a period N

$$F(u,v) = F(u+N,v) = F(u,v+N) = F(u+N,v+N)$$

Proof:

$$F(u,v) = \frac{1}{N} \sum_{x,y=0}^{N-1} \sum f(x,y) e^{-\frac{j2\pi}{N}(ux+vy)}$$

$$F(u+N,v+N) = \frac{1}{N} \sum_{x,y=0}^{N-1} \sum f(x,y) e^{-\frac{j2\pi}{N}(ux+vy+Nx+Ny)}$$

$$F(u+N,v+N) = \frac{1}{N} \sum_{x,y=0}^{N-1} \sum f(x,y) e^{-\frac{j2\pi}{N}(ux+vy)} . e^{-j2\pi(x+y)}$$

$$F(u+N,v+N) = \frac{1}{N} \sum_{x,y=0}^{N-1} \sum f(x,y) e^{-\frac{j2\pi}{N}(ux+vy)}$$

$$F(u+N,v+N) = F(u,v)$$

So we can say that Discrete Fourier Transform is periodic with N

Conjugate

if $f(x,y)$ is a real valued function then

$$\mathbf{F(u,v) = F^*(-u,-v)}$$

Where F^* indicate it complex conjugate

Now Fourier Spectrum

$$|F(u,v)| = |F(-u,-v)|$$

This property help to visualize Fourier Spectrum

Rotation

The rotation property rotates that if a function is rotate by the angle , its fourier transform by an angle amount

Spatial domain

f(x,y) $\longrightarrow$ $\mathbf{f(rcos\,\theta\,,r\sin\theta\,)}$

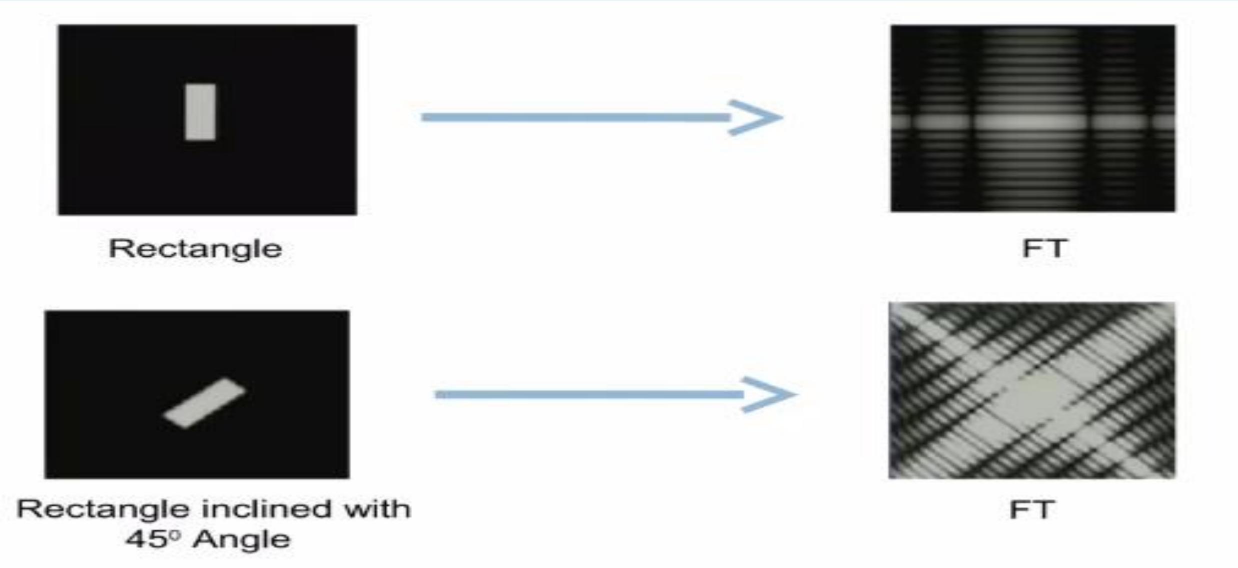
Frequency domain(1-D)

$\mathbf{DFT[f((}\boldsymbol{rcos\,\theta\,,r\,sin\,\theta}\,\mathbf{)]}$ $\longrightarrow$ $\mathbf{F[}\boldsymbol{R}\mathbf{\cos\emptyset\,,}\boldsymbol{R}\mathbf{\sin\emptyset\,]}$

2-D

$\mathbf{DFT[f(rcos(\boldsymbol{\theta}+\boldsymbol{\theta}°),\boldsymbol{r}\sin(\boldsymbol{\theta}+\boldsymbol{\theta}°))]}$ $\longrightarrow$ $\mathbf{F[}\boldsymbol{R\,cos}(\emptyset+\emptyset°)\,,\boldsymbol{R\,sin}(\emptyset+\emptyset°)]$

If the image rotate by a specific angle its spectrum will be rotate



Scaling

Scaling is basically used to increase or decrease the size of image. According to this property, the expansion of a signal in one domain is equal to compression of the signal in one domain is equal to compression of the signal in another domain.

The 2D DFT of a function

f(x,y) is defined as f(x,y) $\longrightarrow$ f(u,v)

If DFT of f(x,y) is f(u,v) then DFT

DFT{f(ax,by)} = $\frac{1}{|ab|} F\left(\frac{u}{a}, \frac{v}{b}\right)$

Distributive

DFT is distributive over addition not on multiplication

$$\mathrm{F}\{f_1(x,y) + f_2(x,y)\} = F\{f_1(x,y)\} + \mathrm{F}\{f_2(x,y)\}$$

$$\mathrm{F}\{f_1(x,y).\, f_2(x,y)\} \neq F\{f_1(x,y)\}.\mathrm{F}\{f_2(x,y)\}$$

Convolution

Convolution is spatial domain is equivalent to multiplication in frequency domain and vice versa

2-D convolution of two array or matrices

$f(x,y)$ & $g(x,y)$ is given by

$DFT\{f(x,y) * g(x,y)\} = F(u,v) \,.\, G(u,v)$

Correlation

Correlation of 2-D signal

$f(x,y) \odot g(x,y) = F^*(u,v) \,.G(u,v)$

$f^*(x,y) \,.\, g(x,y) = F(u,v) \odot G(u,v)$